

PROJECT JÖRMUNGANDR

Executive Summary

Treg Robert Wells

Named AI Collaborator: Corvid

Version 1.0

July 7, 2026

A Public Utility for Ocean Regeneration

Contact:

Email: treg_wells@yahoo.com

Phone: +1 (903) 402-9074

*AI tools were used as research and editing assistance. All concepts, design decisions, and strategic choices are the author's responsibility. The named AI collaborator on this document is **Corvid**. This work draws on publicly available data, peer-reviewed literature, and industry reports.*

1 THE PROBLEM

The ocean is failing.

Acidification, deoxygenation, and plastic pollution are accelerating faster than remediation efforts can respond. The Great Pacific Garbage Patch (GPGP) contains an estimated 45,000–129,000 tons of plastic. Ocean pH has declined from 8.06 to 8.04 in the last decade and continues to drop. Microplastics are now found everywhere—from the surface to the deepest trenches.

Existing solutions are fragmented, nation-state-dependent, and operate on timescales incompatible with ecological collapse. The ocean lacks a governance layer and an infrastructure standards framework comparable to terrestrial systems.

This is not a technology problem. It is an integration problem.

2 THE SOLUTION

Project Jörmungandr is a modular, closed-loop oceanic regeneration architecture designed to restore major oceanic basins through autonomous surface nodes, deep-sea static structures, and mobile mid-water drones.

The architecture operates across four tiers:

Tier	Component	Function
Surface (Passive)	Bio-Bubbles	Algal filtration: CO ₂ drawdown, nutrient uptake, microplastic capture
Surface (Active)	Skin Skimmers	Macroplastic removal from the neuston layer
Mid-Water	Gandr Drones	Active plume suction for microplastic collection
Benthic	SSS Crawlers	Sediment vacuum for heavy metal recovery

The system is designed as a **public utility**—permanent infrastructure for planetary repair, not a business venture. Like a municipal water treatment plant, it is funded by public and philanthropic sources to provide a continuous public good. Cost offsets come from material sales, carbon credits, and a **Sustainability Stamp** licensing program.

3 THE DIFFERENTIATOR: PAP PROTOCOL

What makes Project Jörmungandr unique is the **Polluter Accountability Protocol (PAP)**.

The PAP Protocol:

1. Collects physical samples of pollution
2. Uses **FTIR spectroscopy** to identify polymer signatures
3. Applies **additive fingerprinting** (colorants, flame retardants, stabilizers)
4. Backtracks sources using **ocean current models** (SCUD, HYCOM)
5. Attributes pollution to specific companies or regions
6. Publishes data publicly

No other ocean cleanup project does chemical fingerprinting + backtracking + public attribution at scale. This creates a data product with political and legal utility—not just environmental utility.

4 WHY THIS WORKS

Challenge	Jörmungandr's Answer
Economics of ocean plastic	The Sustainability Stamp sells for brand value, not commodity price—pure margin, no collection costs.
Legal risk	Swiss incorporation, vessel flagging in non-US jurisdictions, \$5–10M legal defense fund.
Nation-state interference	Operates under UNCLOS Article 87 (Freedom of Scientific Research).
Marine mammal entanglement	Comprehensive MMEIA framework: 2 knots trawling speed, acoustic pingers, escape routes, MMOs.
Profit motives	Public Trust governance with Founder's Abdication—zero executive authority after Phase 0.

5 THE PLAN

Project Jörmungandr is deployed in **phased, trigger-based stages**:

Phase	Duration	Focus	Key Activities
Phase 0	3 years	Proof of Concept	Coastal validation of Bio-Bubbles, Gandr, SSS, Skimmers
Phase 1	5 years	Basin-Scale Deployment	Open-ocean deployment in the North Pacific Gyre
Phase 2	10+ years	Global Scaling	Multi-basin deployment, LC-Hive Nuclear

Phase 0 Budget: \$10.4M–\$12.7M

Phase 1 Budget: \$89M–\$136M

Phase 2 Budget (Forecast): \$6.6B–\$13.2B

Note: Phase 2 estimates are conceptual forecasts, not committed budgets. Actual scope and cost will be determined based on Phase 0–1 results.

6 THE COST OFFSET MODEL

Revenue is treated as a **cost offset**, not a profit center. The goal is to reduce the burden on public and philanthropic funding, not to achieve profitability.

Revenue Stream	Phase 1 (Annual)	Phase 2 (Annual)
Recycled Plastics	\$3–8M	\$15–40M
Carbon Credits	\$0.5–3M	\$5–30M
Sustainability Stamp	\$2–10M	\$20–50M
Donations/Grants	\$10–20M	\$25–50M

Target: The project aims to be 70–85% operationally self-sustaining by Phase 2 maturity, with significantly reduced public funding burden.

7 WHAT WE NEED

To begin Phase 0 implementation, we are seeking:

1. **Seed funding** of \$10.4–12.7M to cover:

- Legal incorporation of the Provisional Entity
 - Engineering design and fabrication of Phase 0 systems
 - Strain selection and lab testing
 - Site selection and permitting
2. **Partnerships** with ocean research institutions, legal advocacy organizations, and corporate sustainability programs.
 3. **Diplomatic endorsement** from a host nation to provide diplomatic cover for operations in international waters.

8 WHY NOW

The ocean is at a tipping point. Every year we wait, more plastic enters the system, more CO₂ dissolves into seawater, and more species are lost to entanglement and habitat degradation.

Project Jörmungandr is not a silver bullet. It is an integrated systems architecture that addresses the problem at multiple scales simultaneously—from passive surface filtration to active deep-sea sediment recovery, from physical collection to chemical accountability.

We have the tools, the materials, and the social science today. What we lack is the willingness to connect existing dots into a coherent, circular, and redundant whole. This document is our attempt to draw that integration layer.

"Plant a seed, nurture the tree, share the fruit of your labors."

Document Status	Executive Summary – Ready for Distribution
Date	July 7, 2026
Version	1.0
Author	Treg Robert Wells
Named AI Collaborator	Corvid

*The named AI collaborator on this document is **Corvid**. AI tools were used as research and editing assistance. All concepts, design decisions, and strategic choices are the author's responsibility.*